Q. No.	Questions
UNIT 1	**SECTION 1**
1	A formulation scientist prepares tablets using API, lactose and povidone. During testing, the tablets possess good hardness but fail the disintegration test. Outline the probable reason for the failure. Recommend a suitable excipient and justify your answer. (5 Marks)
2	A pharmaceutical company plans to manufacture chewable tablets for children. The formulation scientist must select a suitable diluent that improves patient acceptability. Based on the scenario provided, recommend a suitable diluent and justify your choice. (5 Marks)
3	During the pre-formulation evaluation of a new oral drug candidate, two formulations exhibited similar drug content and pharmacological activity. However, one formulation dissolved rapidly in gastrointestinal fluids but showed poor membrane permeation, whereas the other crossed biological membranes efficiently despite dissolving slowly. Before finalizing the formulation strategy, the research team decided to evaluate the partition coefficient of both drug candidates. Analyse how the partition coefficient influences the observed behaviour of the two formulations and justify its significance in oral drug delivery. (5 Marks)
4	(i) Two pharmaceutical companies manufacture the same paracetamol tablet. Company A uses only API and lactose, whereas Company B incorporates API, lactose, povidone, starch and magnesium stearate in its formulation. Compare the formulation strategies adopted by the two companies and determine which formulation is likely to produce a pharmaceutically superior tablet. Justify your conclusion. (3 Marks)
	(ii) Based on the above formulation, recommend one additional excipient that Company A should incorporate to improve its tablet formulation and justify your recommendation. (2 Marks)
UNIT 1	**SECTION 2**
5	A pharmaceutical student stated that a biosimilar can be manufactured in the same way as a generic drug because both are developed after the innovator product becomes available. Another student disagreed, explaining that biologics are produced using living systems and therefore cannot be reproduced as exact copies. Evaluate the two opinions by explaining why biosimilars are described as "highly similar" rather than "identical" to the reference biologic. (5 Marks)
6	A pharmaceutical company compared two biosimilar products. • Product A: Identical amino acid sequence; comparable glycosylation profile. • Product B: Identical amino acid sequence; significantly altered glycosylation profile. A pharmaceutical company evaluated two biosimilar candidates during product development. • Product A: Identical amino acid sequence and a glycosylation profile comparable to the reference biologic. • Product B: Identical amino acid sequence but a significantly altered glycosylation profile compared with the reference biologic. The development team proposed that both products should receive regulatory approval because their amino acid sequences are identical. Evaluate the proposal and determine which product is more likely to satisfy biosimilar approval

	requirements. Justify your conclusion based on the critical quality attributes of biosimilar products. (5 Marks)
7	A biosimilar developer successfully confirmed the primary amino acid sequence of a therapeutic protein using advanced analytical techniques. However, the regulatory review committee requested additional evidence before concluding that the biosimilar would perform identically to the reference product during long-term clinical use. Evaluate why analytical characterization alone may not completely eliminate structural uncertainty in biosimilar development. (5 Marks)
8	(i) A pharmaceutical company planned to develop a transdermal patch for systemic drug delivery. During formulation evaluation, scientists emphasized that only molecules possessing suitable physicochemical properties could effectively permeate the skin barrier. Explain the physicochemical requirements necessary for successful transdermal drug delivery. (3 Marks)
	(ii) Based on the above evaluation, justify why Drug B is unsuitable for passive transdermal delivery. (2 Marks)
UNIT 2	**SECTION 1**
9	A researcher observed that a newly developed protein drug showed excellent therapeutic activity in the laboratory but lost its effectiveness during storage. Further investigation revealed exposure to high temperature and repeated shaking during transportation. Predict the factors responsible for the loss of therapeutic activity and explain how formulation development helps maintain the quality of biotherapeutics. (5 Marks)
10	A pharmaceutical company developed two formulations for the same biotherapeutic. Formulation X increased only the apparent solubility of the drug, whereas Formulation Y increased the concentration of free dissolved drug molecules. The development team must choose one formulation for further studies. Evaluate both formulations and predict which formulation would be preferred for improving drug absorption and therapeutic performance. (5 Marks)
11	A biotechnology company is developing two therapeutic candidates for systemic treatment. • Drug X: Molecular weight = 420 Da, Log P = 3.2, Hydrogen bond donors = 2, Hydrogen bond acceptors = 6, stable in gastric fluid. • Drug Y: Molecular weight = 18,000 Da, highly hydrophilic, susceptible to enzymatic degradation, and loses activity on exposure to heat. Despite Drug X satisfying Lipinski's Rule of Five, the company plans an oral tablet, whereas Drug Y is formulated as a lyophilized injectable product. Assess the scientific basis for selecting different formulation approaches for Drug X and Drug Y. (5 Marks)
12	(i) During formulation optimization, two batches of the same biotherapeutic showed identical drug content. However, one batch exhibited better absorption because a different solubility-enhancing excipient had been incorporated. Justify how solubility-enhancing excipients influence the performance of a biotherapeutic formulation. (3 Marks)
	(ii) Based on the above scenario, state two formulation parameters that may improve when drug solubility is enhanced. (2 Marks)
UNIT 2	**SECTION 2**
13	A recombinant protein formulation exhibited visible particles after prolonged storage. Analytical studies confirmed that individual protein molecules had combined to form large complexes, resulting in reduced

	biological activity. The formulation scientist proposed modifying the formulation to improve its shelf life. Predict the cause of reduced therapeutic activity and justify the formulation strategies that can be employed to minimize protein aggregation and enhance shelf life. (5 Marks)
14	During formulation development, a protein solution showed significant loss of active drug even though no chemical degradation was detected. Further investigation revealed that protein molecules had adhered to the walls of the glass container. Predict the reasons for the reduction in active protein concentration and justify the use of surfactants in improving the shelf life of protein formulations. (5 Marks)
15	A vaccine formulation was accidentally exposed to freezing temperatures during transportation. After thawing, visible particles and aggregation of the vaccine protein were observed. Recommend a suitable excipient to minimize freezing-induced aggregation in future batches and explain its protective mechanism. (5 Marks)
16	(i) During stability studies, a protein biotherapeutic exhibited a gradual loss of biological activity due to oxidative degradation. Analyze the effect of oxidative degradation on the biological activity and therapeutic performance of a protein biotherapeutic formulation. (3 Marks)
	(ii) Select an appropriate formulation approach to improve the stability of the protein biotherapeutic and justify your choice. (2 Marks)
UNIT 3	**SECTION 1**
17	A pharmaceutical company assigned the following suitability ratings for different dosage forms as provided in the below figure 'Topical formulations comparison chart".  Figure: Topical formulations comparison chart Using the matrix, rank the four dosage forms from the most suitable to the least suitable for managing dry chronic skin conditions.Justify your ranking. (5 Marks)
18	During stability testing, two liquid formulations failed quality evaluation. • Product A: A permanent solid mass formed at the bottom of the container and could not be redispersed by shaking. • Product B: Two distinct liquid layers remained permanently separated after storage. Analyse the root cause of each failure and explain how the observed instability affects the pharmaceutical quality of the respective formulation.

(5 Marks)

19 A quality review committee made the following observations after evaluating four liquid dosage forms.

- Observation I: Solutions provide rapid absorption because the active pharmaceutical ingredient (API) is already dissolved.
- Observation II: Syrups are naturally self-preserving because of their high osmotic pressure.
- Observation III: Flocculated suspensions settle slowly and form irreversible caking.
- Observation IV: Oil-in-water emulsions are preferred for the oral delivery of unpalatable oily drugs.
- Observation V: Suspensions are preferred only when the drug is completely soluble in the dispersion medium.

Evaluate the observations and determine which require revision before the committee report can be approved. Justify your decision.

(5 Marks)

20 (i)

Excipient Category	**Recorded Function**
Binder/Adhesive	Promotes particle adhesion
Disintegrant	Improves powder flow during compression
Lubricant/Glidant	Prevents sticking to punches

Review the formulation record as provided in the above table and identify the incorrectly assigned function. Justify your answer.

(3 Marks)

(ii) Based on the audit findings, identify the excipient category that requires functional correction before formulation approval.

(2 Marks)